

Q 1

ACE

What is the value of $\sqrt{\frac{(3.059)^4}{16.38-11.063}}$ correct to three significant figures?

- (A) 1.759
(B) 1.76
(C) 4.058
(D) 4.06

$$\sqrt{\frac{(3.059)^4}{16.38-11.063}} = 4.058126902... \approx 4.06 \quad \boxed{D}$$

(Correct to 3 significant figures)

Q 2

ACE

Express $0.005\dot{6}$ as a fraction in its simplest form.

Let $x = 0.005\dot{6}$
 $x = 0.0056666... \quad (1)$
 $10x = 0.0566666... \quad (2)$
 Eqn (2)-(1)
 $9x = 0.051$
 $x = \frac{0.051}{9} = \frac{51}{9000} = \frac{17}{3000}$

Q 3

ACE

What is the value of $\frac{(4.13)^2 - 12.78}{\sqrt{16.65 + 9.31 \times 1.75}}$ correct to three significant figures?

- (A) 0.209
(B) 0.210
(C) 0.745
(D) 0.746

$$\frac{(4.13)^2 - 12.78}{\sqrt{16.65 + 9.31 \times 1.75}} = 0.7451622041... \approx 0.745 \text{ (3 significant figures)} \quad \boxed{C}$$

Q 4

ACE

The base length x , of a square pyramid of volume V and perpendicular height h , is given by the formula $x = \sqrt{\frac{3V}{h}}$. Find the value of x , if $V = 750$ and $h = 6.95$. Answer correct to 2 decimal places

$$x = \sqrt{\frac{3V}{h}} = \sqrt{\frac{3 \times 750}{6.95}} = 17.99280432... \approx 17.99$$

Q 5

ACE

Simplify $\frac{2^n \times 4^{n+1}}{8^n}$.

$$\frac{2^n \times 4^{n+1}}{8^n} = \frac{2^n \times 2^{2n+2}}{2^{3n}} = 2^2 = 4$$

Q 6

ASCHAM

The number of nurses on the staff of a hospital is reduced by 12%. There are now 264 nurses. How many nurses were there originally?

$$\begin{aligned} 88\% &\rightarrow 264 \\ 1\% &\rightarrow \frac{264}{88} \\ 100\% &\rightarrow 100 \times \frac{264}{88} \\ &= 300 \\ \therefore \text{there were 300 nurse} \end{aligned}$$



Q 7

ASCHAM

Expand and simplify $4m^{-2} \div \frac{1}{2}m^{-1}$

$$\begin{aligned} 4m^{-2} \div \frac{1}{2}m^{-1} \\ = 8m^{-2-(-1)} \\ = \frac{8}{m} \end{aligned}$$

Q 8

ASCHAM

Simplify $\frac{16^{n+2} \times 4^{n-1}}{64^n}$

$$\begin{aligned} \frac{16^{n+2} \times 4^{n-1}}{64^n} &= \frac{(4^2)^{n+2} \times 4^{n-1}}{4^{3n}} \\ &= \frac{4^{2n+4} \times 4^{n-1}}{4^{3n}} \\ &= \frac{4^{3n+3}}{4^{3n}} \\ &= 4^3 \\ &= 64 \end{aligned}$$

Q 9

BLAKEHURST

At Octopus Communications' annual sale, all mobile phones were discounted by 40%. Cedric paid \$630 for a mobile phone at the sale. What was the original price of the phone?

$$60\% = \$630$$

$$1\% = 10.5$$

$$100\% = \$1050$$

Q10

BLAKEHURST

Simplify $\frac{16}{2^{3x} \times 8^{1-x}}$

$$\frac{2^4}{2^{3x} \times 2^{3(1-x)}}$$

$$\frac{2^4}{2^{3x} \times 2^{3-3x}}$$

$$\frac{2^4}{2^3} = 2$$

Q11

If $x = 3^2$ and $y = 2^4$, express $27^6 \times 4^8$ in terms of x and y

DET

$$\begin{aligned} & (3^3)^6 \times (2^2)^8 \\ &= 3^{18} \times 2^{16} \\ &= (3^2)^9 \times (2^4)^4 \\ &= x^9 \times y^4 \end{aligned}$$

Q12

Simplify without negative indices $\frac{1+x^{-1}}{x-x^{-1}}$

FORT ST

$$\begin{aligned} \frac{1+x^{-1}}{x-x^{-1}} &= \frac{1+\frac{1}{x}}{x-\frac{1}{x}} \\ &= \frac{\frac{x+1}{x}}{\frac{x^2-1}{x}} \\ &= \frac{x+1}{x^2-1} \\ &= \frac{x+1}{x} \times \frac{x}{x^2-1} \\ &= \frac{x+1}{x^2-1} \\ &= \frac{x+1}{(x+1)(x-1)} \\ &= \frac{1}{x-1} \end{aligned}$$

Q13

A country has a 15% rate of GST. A golf club sells for \$184 which includes GST. Find the amount of GST paid for the golf club.

GOSFORD

$$\begin{aligned} 115\% &= 184 \\ 1\% &= 184 \div 115 \\ 100\% &= (184 \div 115) \times 100 \\ &= \$160 \quad \therefore \text{GST} = \$24 \end{aligned}$$

Q14

A pair of jeans were discounted by 15% to a selling price of \$63.75. Find the original marked price of the jeans before the discount was applied.

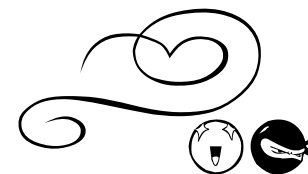
GOSFORD

$$\begin{aligned} 85\% &= \$63.75 \\ 1\% &= \$63.75 \div 85 = \$0.75 \\ 100\% &= \$0.75 \times 100 \\ &= \$75 \quad \textcircled{2} \\ \therefore \text{Cost} &= \$75 \end{aligned}$$



BASIC ARITHMETIC

PART A: numbers



Q15

What is 6.0951 correct to three significant figures?

KINGS

- (A) 6.09
(B) 6.10
(C) 6.096
(D) 6.0956

Q16

Express in simplest form $\sqrt{9a^{16}}$

KINGS

$$3a^8$$

Q17

The earth revolves around the sun at a speed of approximately 106560 km/h. How far does the earth travel in a year (365 days)? Answer in scientific notation correct to 4 significant figures.

MERIDAN

$$\begin{aligned} D &= S \times T \\ &= 106560 \times 365 \times 24 \\ &= 933465600 \\ &= 9.335 \times 10^8 \text{ km} \end{aligned}$$

Q18

Express with positive indices $\frac{a^{-1} + b^{-1}}{(ab)^{-1}}$

MERIDAN

$$\begin{aligned} &\left(\frac{1}{a} + \frac{1}{b}\right) \div \frac{1}{ab} \\ &= \frac{b+a}{ab} \times \frac{ab}{1} \\ &= b+a \end{aligned}$$

Q19

The sale price of a TV set after $3\frac{1}{2}\%$ discount is \$1929. What is the original cost of the TV? (correct to the nearest dollar)

MERIDAN

$$\begin{aligned} &\frac{1929 \times 100}{96.5} \\ &= 1998.96 \dots \\ &= \$1999 \text{ (nearest \$)} \end{aligned}$$

Q20

Simplify $\frac{(6x^3y)^2}{6x^2y^4}$.

NSGHS

$$\begin{aligned} \frac{(6x^3y)^2}{6x^2y^4} &= \frac{6^2x^6y^2}{6x^2y^4} \\ &= \frac{6x^4}{y^2} \end{aligned}$$

Q21

Express $\frac{x}{y}$ without negative indices if $x = a^{-2}b^3$ and $y = a^5b^{-2}$.

NSGHS

$$\begin{aligned} \frac{x}{y} &= \frac{a^{-2}b^3}{a^5b^{-2}} \\ &= \frac{b^5}{a^7} \end{aligned}$$

Q22

Jake's drum kit cost him \$2430 which was a 8% increase from the previous years price. What did the drum kit sell for the previous year?

QAT

Answer

108% of previous price = \$2430

1% of previous price = \$22.50

100% of previous price = \$2250

∴ the previous year price was \$2250

Q23

The crowd attendance at a football game on the weekend was 15 660 which was a decline of 10% from the previous weekend. How many people attended the previous weekend?

SCEGGS

90% = 15660 people ✓

10% = 1740 people

100% = 17400 people

17400 attended the previous weekend ✓

Q24

Divide the product of 2.9384×10^{12} and 1.58×10^{11} by 2.068×10^{10}
Give your answer correct to 3 significant figures

SYD TECH

$$2.25 \times 10^{13}$$

Q25

Brian pays \$34.50 for a shirt which was originally marked at \$45.00

Calculate the percentage discount correct to one decimal place

SYD TECH

$$\text{Discount} = \$10.50$$

$$\frac{10.50}{45} \times \frac{100}{1} = 23.3\%$$

Q26

The cost of a calculator is \$16 after it was reduced by 4.5%. What was the original price?

ALL
SAINTS

$$(100 - 4.5)\% = 16$$

$$95.5\% = 16$$

$$1 = 0.167539...$$

$$100\% = \$16.75$$

Q27

Rewrite the expression $5(x+7)^{-2}$ without fractional index

ALL
SAINTS

$$\frac{5}{\sqrt[5]{(x+7)^2}}$$

Q28

How much time will it take for a rocket travelling at a constant speed of 1.5×10^4 km/h to reach the Moon, if the distance to the Moon is 3.84×10^5 km?

ALL
SAINTS

$$t = \frac{\text{distance}}{\text{velocity}}$$

$$= \frac{3.84 \times 10^5}{1.5 \times 10^4}$$

$$= 25.6h$$



Q 1

ACE

What are the values of a and b if $\sqrt{50} - \sqrt{72} = a\sqrt{b}$?

- (A) $a = -11$ and $b = 2$
 (B) $a = -1$ and $b = 2$
 (C) $a = -11$ and $b = 22$
 (D) $a = -1$ and $b = 22$

$$\begin{aligned}\sqrt{50} - \sqrt{72} &= 5\sqrt{2} - 6\sqrt{2} \\ &= -1\sqrt{2}\end{aligned}$$

B

Solution is $a = -1$ and $b = 2$.

Q 2

ACE

Express $\frac{2}{3+\sqrt{10}}$ as a fraction with a rational denominator,

$$\begin{aligned}\frac{2}{3+\sqrt{10}} &= \frac{2}{3+\sqrt{10}} \times \frac{3-\sqrt{10}}{3-\sqrt{10}} \\ &= \frac{2(3-\sqrt{10})}{9-10} \\ &= 2(\sqrt{10}-3)\end{aligned}$$

Q 3

ACE

Simplify $\sqrt{32} - \sqrt{18} + \sqrt{2}$.

$$\begin{aligned}\sqrt{32} - \sqrt{18} + \sqrt{2} &= 4\sqrt{2} - 3\sqrt{2} + \sqrt{2} \\ &= 2\sqrt{2}\end{aligned}$$

Q 4

ASCHAM

What is $\frac{4\sqrt{3}}{\sqrt{3}-\sqrt{2}} + \frac{4\sqrt{3}}{\sqrt{3}+\sqrt{2}}$ expressed with a rational denominator?

- (A) 24
 (B) 48
 (C) 72
 (D) 144

$$\begin{aligned}\frac{4\sqrt{3}}{\sqrt{3}-\sqrt{2}} + \frac{4\sqrt{3}}{\sqrt{3}+\sqrt{2}} &= \frac{4\sqrt{3}(\sqrt{3}+\sqrt{2}) + 4\sqrt{3}(\sqrt{3}-\sqrt{2})}{(\sqrt{3}-\sqrt{2})(\sqrt{3}+\sqrt{2})} \\ &= \frac{12 + 4\sqrt{6} + 12 - 4\sqrt{6}}{3-2} = 24\end{aligned}$$

A

Q 5

BAULKHAM

If $\frac{2\sqrt{2}}{2\sqrt{2}-3} = a - \sqrt{b}$ then the values of a and b are:

- (A) -4 and 12
 (B) -4 and 72
 (C) -8 and 12
 (D) -8 and 72

Q 6

DET

If $\sqrt{48} + 2\sqrt{12} = 8\sqrt{2a+2}$, find the value of a

$$\begin{aligned}\sqrt{48} + 2\sqrt{12} &= 4\sqrt{3} + 4\sqrt{3} = 8\sqrt{3} \\ \text{but } \sqrt{48} + 2\sqrt{12} &= 8\sqrt{2a+2}, \\ \text{so } 8\sqrt{2a+2} &= 8\sqrt{3} \\ \therefore 2a+2 &= 3, 2a=1, \\ a &= \frac{1}{2}\end{aligned}$$

Q 7

FORT ST

Find the values of a and b if $\frac{5}{1+\sqrt{5}} = a + b\sqrt{5}$

$$\begin{aligned}\frac{5}{1+\sqrt{5}} &= \frac{5}{1+\sqrt{5}} \times \frac{1-\sqrt{5}}{1-\sqrt{5}} \\ &= \frac{5-5\sqrt{5}}{1-5} \\ &= \frac{5-5\sqrt{5}}{-4} \\ &= -\frac{5}{4} + \frac{5}{4}\sqrt{5} \\ \Rightarrow a &= -\frac{5}{4}, b = \frac{5}{4}\end{aligned}$$

Q 8

GIRAWEEEN

Show that $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ is a rational number.

$$\begin{aligned}1) \quad &\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}} \\ &= \frac{3+\sqrt{2} + 3-\sqrt{2}}{(3-\sqrt{2})(3+\sqrt{2})} \\ &= \frac{6}{7}\end{aligned}$$

Q 9

FREEMAN

Simplify as a single fraction with a rational denominator $\frac{1}{\sqrt{3}} + \frac{1}{3+\sqrt{3}}$.

$$\begin{aligned}&\frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} + \frac{1}{3+\sqrt{3}} \times \frac{3-\sqrt{3}}{3-\sqrt{3}} \\ &= \frac{\sqrt{3}}{3} + \frac{3-\sqrt{3}}{6} \\ &= \frac{2\sqrt{3} + 3 - \sqrt{3}}{6} \\ &= \frac{\sqrt{3} + 3}{6}\end{aligned}$$

Q10

GIRAWEEEN

Find the value of the integers a, b and c if:

$$\frac{6}{2\sqrt{7}-5} = a + b\sqrt{c}$$

$$\begin{aligned}\frac{6}{2\sqrt{7}-5} &= \frac{6}{2\sqrt{7}-5} \times \frac{2\sqrt{7}+5}{2\sqrt{7}+5} \\ &= \frac{6(2\sqrt{7}+5)}{(2\sqrt{7}-5)(2\sqrt{7}+5)} \\ &= \frac{12\sqrt{7} + 30}{(2\sqrt{7})^2 - 5^2} = \frac{12\sqrt{7} + 30}{28 - 25} \\ &= \frac{12\sqrt{7} + 30}{3} \\ &= 4\sqrt{7} + 10 \quad \text{③} \\ &= 10 + 4\sqrt{7} \\ a &= 10, b = 4, c = 7\end{aligned}$$

Q11

GIRAWEEEN

Rationalise the denominator: $\frac{\sqrt{5}-\sqrt{2}}{3\sqrt{2}-\sqrt{5}}$

$$\begin{aligned}\frac{\sqrt{5}-\sqrt{2}}{3\sqrt{2}-\sqrt{5}} &= \frac{(\sqrt{5}-\sqrt{2})(3\sqrt{2}+\sqrt{5})}{(3\sqrt{2}-\sqrt{5})(3\sqrt{2}+\sqrt{5})} \\ &= \frac{3\sqrt{10}+5-6-\sqrt{10}}{(3\sqrt{2})^2-(\sqrt{5})^2} \\ &= \frac{2\sqrt{10}-1}{18-5} = \frac{2\sqrt{10}-1}{13}\end{aligned}$$

Q12

GIRAWEEEN

By expressing $\frac{4}{2+\sqrt{5}} - \frac{1}{9-4\sqrt{5}}$ in its simplest form show that it is a rational number.

$$\begin{aligned}&\frac{4}{2+\sqrt{5}} - \frac{1}{9-4\sqrt{5}} \\ &= \frac{4(9-4\sqrt{5}) - (2+\sqrt{5})}{(2+\sqrt{5})(9-4\sqrt{5})} \\ &= \frac{34-17\sqrt{5}}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} \\ &= \frac{34\sqrt{5}+68-85-34\sqrt{5}}{5-4} \\ &= -17\end{aligned}$$

Q13

GIRAWEEEN

Simplify: $(\sqrt{2}+1)^2 + (2\sqrt{3})^2$

$$\begin{aligned}&(\sqrt{2}+1)^2 + (2\sqrt{3})^2 \\ &= 2 + 2\sqrt{2} + 1 + 12 \\ &= 15 + 2\sqrt{2}\end{aligned}$$

Q14

GIRAWEEEN

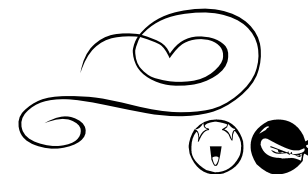
Find the values of a and b if $\frac{\sqrt{6}+2}{\sqrt{6}-2} = a + \sqrt{b}$

$$\begin{aligned}&b) \frac{\sqrt{6}+2}{\sqrt{6}-2} \times \frac{\sqrt{6}+2}{\sqrt{6}+2} \\ &= \frac{(\sqrt{6}+2)^2}{6-4} \\ &= \frac{10+4\sqrt{6}}{2} \\ &= 5+2\sqrt{6} \\ &= 5+\sqrt{24} \\ \therefore a &= 5 \quad b = 24.\end{aligned}$$



BASIC ARITHMETIC

PART B: SURDS



Q15

If $x = 7 - 4\sqrt{3}$, show that $x + x^{-1}$ is rational, and find its value.

MERIDAN

$$\begin{aligned}
 & x + x^{-1} \\
 & 7 - 4\sqrt{3} + \frac{1}{7 - 4\sqrt{3}} \\
 & 7 - 4\sqrt{3} + \frac{1}{7 - 4\sqrt{3}} \times \frac{7 + 4\sqrt{3}}{7 + 4\sqrt{3}} \\
 & \frac{7 - 4\sqrt{3} + 7 + 4\sqrt{3}}{49 - 16 \times 3} \\
 & = \frac{14}{1} \\
 & = 14
 \end{aligned}$$

Q16

If $\sqrt{56} + \sqrt{14} = \sqrt{A}$, find A .

GIRAWEEEN

$$\begin{aligned}
 \sqrt{56} + \sqrt{14} &= \sqrt{4 \times 14} + \sqrt{14} \\
 &= 2\sqrt{14} + \sqrt{14} \\
 &= 3\sqrt{14} \\
 &= \sqrt{9 \times 14} \\
 &= \sqrt{126} \\
 A &= 126
 \end{aligned}$$

Q17

Find the values of a and b if $\frac{3}{\sqrt{5}-2} = \sqrt{a} + b$.

SCEGGS

$$\begin{aligned}
 \frac{3}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} &= \frac{3\sqrt{5}+6}{5-4} \\
 &= \sqrt{45} + 6 \\
 \therefore a &= 45, b = 6
 \end{aligned}$$

Q18

Find a and b if $\frac{\sqrt{7}-5}{-2+3\sqrt{7}} = a + b\sqrt{7}$

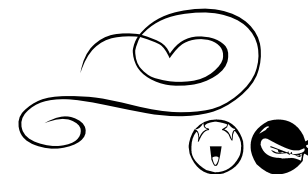
SYD TECH

$$\begin{aligned}
 \frac{\sqrt{7}-5}{3\sqrt{7}-2} \times \frac{3\sqrt{7}+2}{3\sqrt{7}+2} &= \frac{21+2\sqrt{7}-15\sqrt{7}-10}{63-4} \\
 &= \frac{11-13\sqrt{7}}{59} \\
 a &= \frac{11}{59}, b = -\frac{13}{59}
 \end{aligned}$$



BASIC ARITHMETIC

PART B: SURDS



Q19

The exact value of $k^4 - 3k^2 + 2$ when $k = 2\sqrt{2}$ is:

ALL SAINTS

- (a) $12\sqrt{2}$
- (b) $4\sqrt{2} + 2$
- (c) 42
- (d) 54

Q20

$\frac{5 - 2\sqrt{3}}{2 - \sqrt{3}}$ Can be expressed in the form $a + b\sqrt{3}$

WESTERN

Find the value of a and b .

$$\begin{aligned} & \textcircled{b} \frac{(5 - 2\sqrt{3}) \times (2 + \sqrt{3})}{(2 - \sqrt{3}) \times (2 + \sqrt{3})} \\ & = \frac{10 + 5\sqrt{3} - 4\sqrt{3} - 6}{2^2 - (\sqrt{3})^2} \\ & = \frac{4 + \sqrt{3}}{4 - 3} \\ & = 4 + \sqrt{3} \\ & a = 4, b = 1 \checkmark \end{aligned}$$

Q21

Simplify: $\frac{2}{\sqrt{6}} + \frac{1}{3 + \sqrt{6}}$

SCEGGS

$$\begin{aligned} \frac{2\sqrt{6}}{6} + \frac{3 - \sqrt{6}}{9 - 6} &= \frac{\sqrt{6}}{3} + \frac{3 - \sqrt{6}}{3} \\ &= \frac{\sqrt{6} + 3 - \sqrt{6}}{3} = \frac{3}{3} = 1 \end{aligned}$$

Q22

Simplify $\sqrt{24} + 3\sqrt{48} - 2\sqrt{54} + 4\sqrt{75}$, leaving your answer in the simplest surd form.

SYD TECH

$$\begin{aligned} & 2\sqrt{6} + 12\sqrt{3} - 6\sqrt{6} + 20\sqrt{3} \\ & = 32\sqrt{3} - 4\sqrt{6} \end{aligned}$$



BASIC ARITHMETIC

PART C: FACTORISING

Unit

SELECTIVE

Q 1

Simplify $\frac{2x^3+128}{x^2-9} \div \frac{x+4}{x-3}$.

ACE

$$\begin{aligned}\frac{2x^3+128}{x^2-9} \div \frac{x+4}{x-3} &= \frac{2(x^3+64)}{(x+3)(x-3)} \times \frac{(x-3)}{(x+4)} \\ &= \frac{2(x+4)(x^2-4x+16)}{(x+3)(x-3)} \times \frac{(x-3)}{(x+4)} \\ &= \frac{2(x^2-4x+16)}{(x+3)}\end{aligned}$$

Q 2

Factorise completely $9a^2 - b^2 + 3a - b$.

ASCHAM

$$\begin{aligned}9a^2 - b^2 + 3a - b &= (3a-b)(3a+b) + (3a-b) \\ &= (3a-b)(3a+b+1)\end{aligned}$$

Q 3

Factorise completely:

(i) $y^2 - 6y - 16$

(ii) $(x-2)^2 - 16$

$$y^2 - 6y - 16 = (y-8)(y+2)$$

$$\begin{aligned}(x-2)^2 - 16 &= (x-2)^2 - 4^2 \\ &= ((x-2)+4)((x-2)-4) \\ &= (x+2)(x-6)\end{aligned}$$

Q 4

Simplify $\frac{x^3-1}{x^2-1} \times \frac{x^2-4x-5}{4x^2+4x+4}$

BAULKHAM

$$\begin{aligned}\frac{x^3-1}{x^2-1} \times \frac{x^2-4x-5}{4x^2+4x+4} &= \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} \times \frac{(x+1)(x-5)}{4(x^2+x+1)} \\ &= \frac{(x-5)}{4}\end{aligned}$$

Q 5

Completely simplify

BRIGIDINE

$$\frac{x^2-9}{x^2-x-12} \times \frac{x^2-9x+20}{x^2-3x}$$

$$\begin{aligned}&\frac{(x-3)(x+3)}{(x-4)(x+3)} \times \frac{(x-4)(x-5)}{x(x-3)} \\ &= \frac{x-5}{x}\end{aligned}$$





BASIC ARITHMETIC

PART C: FACTORISING

Unit

SELECTIVE

Q 6

Completely simplify

BRIGIDINE

$$\frac{5}{a^2 - 3a - 4} - \frac{3}{a^2 - a - 2}$$

$$\begin{aligned} & \frac{5}{(a+1)(a-4)} - \frac{3}{(a+1)(a-2)} \\ &= \frac{5(a-2) - 3(a-4)}{(a+1)(a-4)(a-2)} \\ &= \frac{5a - 10 - 3a + 12}{(a+1)(a-4)(a-2)} \\ &= \frac{2a + 2}{(a+1)(a-4)(a-2)} \\ &= \frac{2(a+1)}{(a+1)(a-4)(a-2)} \\ &= \frac{2}{(a-4)(a-2)} \end{aligned}$$

Q 7

DET

Simplify $\frac{m^3 + m^2}{x^2 - x} \div \frac{m+1}{x - x^3}$ as a single fraction in simplest form

$$\begin{aligned} & \frac{m^3 + m^2}{x^2 - x} \div \frac{m+1}{x - x^3} \\ &= \frac{m^2(m+1)}{x(x-1)} \times \frac{-x(x^2-1)}{m+1} \\ &= \frac{m^2(\cancel{m+1})}{x(x-1)} \times \frac{-x(x+1)(\cancel{x-1})}{\cancel{m+1}} \\ &= -m^2(x+1) \end{aligned}$$

Q 8

Factorise $6a^3 - 48$.

FORT ST

$$6(a^3 - 8) = 6(a-2)(a^2 + 2a + 4)$$

Q 9

Factorise fully: $5x^2 - 2x - 3$

GIRAWREEN

$$\begin{aligned} & (b)(i) \quad 5x^2 - 2x - 3 \\ &= (x-1)(5x+3) \quad (2) \end{aligned}$$

Q10

Factorise fully: $x^4 - y^4$

GIRAWREEN

$$\begin{aligned} & x^4 - y^4 \\ &= (x^2)^2 - (y^2)^2 \\ &= (x^2 + y^2)(x^2 - y^2) \quad (3) \\ &= (x^2 + y^2)(x+y)(x-y) \end{aligned}$$

Q11

Express $\frac{x-6}{x^2+4} + \frac{x+2}{x^2-4}$ as a single fraction and simplify.

GIRAWEEEN

$$\begin{aligned}
 & b) \frac{x-6}{x^2+4} + \frac{x+2}{x^2-4} \\
 & = \frac{x-6}{x^2+4} + \frac{x+2}{(x-2)(x+2)} \\
 & = \frac{(x-6)(x-2) + (x^2+4)}{(x^2+4)(x-2)} \\
 & = \frac{x^2 - 8x + 12 + x^2 + 4}{(x^2+4)(x-2)} \\
 & = \frac{2x^2 - 8x + 16}{(x^2+4)(x-2)}
 \end{aligned}$$

Q12

Factorise fully: $4x^2 + 16x - 9$

GIRAWEEEN

$$\begin{aligned}
 & 4x^2 + 16x - 9 \\
 & (4x^2 - 2x) + (18x - 9) \\
 & 2x(2x - 1) + 9(2x - 1) \\
 & (2x + 9)(2x - 1)
 \end{aligned}$$

Q13

Factorise fully: $27 - y^3$

GIRAWEEEN

$$\begin{aligned}
 & 27 - y^3 \\
 & (3 - y)(9 + 3y + y^2)
 \end{aligned}$$

Q14

Factorise fully: $x^3 - 2x^2 - 15x$

FORT ST

$$\begin{aligned}
 & i) \frac{x^3 - 2x^2 - 15x}{x^3 - 2x^2 - 15x} \\
 & = x(x^2 - 2x - 15) \\
 & = x(x - 5)(x + 3)
 \end{aligned}$$

Q15

Factorise fully: $x^3 + 8$

GIRAWEEEN

$$\begin{aligned}
 & x^3 + 8 \\
 & = (x + 2)(x^2 - 2x + 4)
 \end{aligned}$$

Q16

Express as a single fraction, leaving the denominator in factorised form:

$$\frac{x+5}{x^2+x} + \frac{x+4}{x^2-x}$$

GIRAWEEEN

$$\begin{aligned}
 & \frac{x+5}{x^2+x} + \frac{x+4}{x^2-x} \\
 & = \frac{(x+5)(x-1) + (x+4)(x+1)}{x(x+1)(x-1)} \\
 & = \frac{x^2 + 4x - 5 + x^2 + 5x + 4}{x(x+1)(x-1)} \\
 & = \frac{2x^2 + 9x - 1}{x(x+1)(x-1)} \quad \textcircled{3}
 \end{aligned}$$



BASIC ARITHMETIC

PART C: FACTORISING

2unit

SELECTIVE

Q17

Simplify fully: $\frac{x^2 + y^2 + 2xy}{x^2 - y^2}$

GOSFORD

$$\begin{aligned} & \frac{x^2 + y^2 + 2xy}{x^2 - y^2} \\ &= \frac{(x+y)^2}{(x-y)(x+y)} \\ &= \frac{x+y}{x-y} \end{aligned}$$

Q18

Factorise $x^3 - 8y^3$

GIRAWEEEN

$$x^3 - (2y)^3 = (x - 2y)(x^2 + 2xy + 4y^2)$$

Q19

Simplify fully $\frac{5a - a^2}{a^2 - 4a - 5}$

KINGS

$$\begin{aligned} & \frac{a(5-a)}{a^2 - 4a - 5} = \frac{-a(a-5)}{(a-5)(a+1)} \\ &= -\frac{a}{a+1} \end{aligned}$$

Q20

Factorise completely: $8x^3 + 125y^3$

KINGS

$$8x^3 + 125y^3 = (2x + 3y)(4x^2 - 6xy + 9y^2)$$

(Correct substitution of formula)

Q21

Factorise completely: $-6x^2 - 11x + 10$

KINGS

$$\begin{aligned} & -6x^2 - 11x + 10 \\ &= -6x^2 - 15x + 4x + 10 \\ &= -3x(2x + 5) + 2(2x + 5) \\ &= (2x + 5)(-3x + 2) \text{ or } 4x \\ &= -(3x - 2)(2x + 5) \end{aligned}$$

Q22

Factorise completely: $x^2y - 2x^2 - 4y + 8$

KINGS

$$\begin{aligned} & x^2y - 2x^2 - 4y + 8 \\ &= x^2(y - 2) - 4(y - 2) \\ &= (x^2 - 4)(y - 2) \\ &= (y - 2)(x - 2)(x + 2) \end{aligned}$$



BASIC ARITHMETIC

PART C: FACTORISING

2unit

SELECTIVE

Q23

Simplify completely: $\frac{2}{x-1} - \frac{x+2}{x^2-2x+1}$

KINGS

$$\begin{aligned} \frac{2}{x-1} - \frac{x+2}{x^2-2x+1} &= \frac{2}{x-1} - \frac{x+2}{(x-1)(x-1)} \\ &= \frac{2(x-1) - (x+2)}{(x-1)^2} \\ &= \frac{2x-2-x-2}{(x-1)^2} \\ &= \frac{x-4}{(x-1)^2} \end{aligned}$$

Q24

Simplify fully: $\frac{2}{x^2+8x+12} + \frac{1}{x^2-3x-10}$

KINGS

$$\begin{aligned} &\frac{2}{(x+6)(x+2)} + \frac{1}{(x-5)(x+2)} \\ &= \frac{2(x-5) + (x+6)}{(x+6)(x+2)(x-5)} \\ &= \frac{3x-4}{(x+6)(x+2)(x-5)} \end{aligned}$$

Q25

Solve the equation $|2x+1| = 3x-2$

KINGS

$$\begin{aligned} \text{Case 1: } 2x+1 &= 3x-2 \quad \text{Case 2: } 2x+1 = -(3x-2) \\ x &= 3 \quad \checkmark \quad 2x+1 = -3x+2 \\ \text{Sub } x=3 & \quad \checkmark \quad 6x=1 \quad x=\frac{1}{6} \\ |2(3)+1| &= 3(3)-2 \quad \text{Sub } x=\frac{1}{6} \\ |7| &= 7 \quad |2(\frac{1}{6})+1| \neq 3(\frac{1}{6})-2 \\ \text{Not a Soln} & \quad \checkmark \quad \therefore x=3 \end{aligned}$$

Q26

Simplify completely $\frac{z^2+4z}{z^2+3z-4} \div \frac{8z}{z^2-1}$

KINGS

$$\begin{aligned} &\frac{z(z+4)}{(z+4)(z-1)} \times \frac{(z+1)(z-1)}{8z} \\ &= \frac{z+1}{8} \quad \checkmark \end{aligned}$$

Q27

Factorise fully, x^4+125x

KNOX

$$\begin{aligned} x^4+125x &= x(x^3+125) \\ &= x(x+5)(x^2-5x+25) \end{aligned}$$





BASIC ARITHMETIC

PART C: FACTORISING

2unit

SELECTIVE

Q28

Factorise fully, $100 - (40 - t)^2$

KNOX

$$\begin{array}{l} (11) \quad 100 - (40 - t)^2 \\ \underline{(10 - 40 + t)(10 + 40 - t)} \\ (t - 30)(50 - t) \quad \checkmark \end{array}$$

Q29

Simplify $\frac{(a + b)^3 - a^3}{b}$

MERIDAN

$$\begin{array}{l} \frac{(a+b)^3 - a^3}{b} \\ \downarrow \\ \frac{[(a+b) - a][a^2 + a(a+b) + a^2]}{b} \\ \downarrow \\ \frac{[a+b-a][a^2 + 2ab + b^2 + a^2 + ab + a^2]}{b} \\ \downarrow \\ \frac{b[3a^2 + 3ab + b^2]}{b} \\ \downarrow \\ 3a^2 + 3ab + b^2 \end{array}$$

Q30

Fully simplify the expression: $\frac{x^3 + 1}{x^2 - x - 6} \div \frac{x^2 - x + 1}{x^2 - 9}$

ALL
SAINTS

$$\begin{aligned} \frac{x^3 + 1}{x^2 - x - 6} \div \frac{x^2 - x + 1}{x^2 - 9} &= \frac{(x+1)(x^2 - x + 1)}{(x-3)(x+2)} \times \frac{(x+3)(x-3)}{x^2 - x + 1} \\ &= \frac{(x+1)(x+3)}{(x+2)} \end{aligned}$$

Q31

Simplify $\frac{m^2 - 25}{m^2 - 7m + 10} \div \frac{m^2 + 9m + 20}{m^2 + m - 6}$

WESTERN

$$\begin{array}{l} \frac{(m+5)(m-5)}{(m+5)(m-2)} \times \frac{(m+3)(m-2)}{(m+4)(m+5)} \quad \checkmark \\ \hline \frac{m+3}{m+4} \quad \checkmark \end{array}$$

Q 1

Solve $\frac{x-5}{3} - \frac{x+1}{4} = 2$.

ACE

$$\frac{x-5}{3} - \frac{x+1}{4} = 2$$

$$\cancel{12} \times \left(\frac{x-5}{\cancel{3}} \right) - \left(\frac{x+1}{\cancel{4}} \right) \times \cancel{12}^3 = 2 \times 12$$

$$4x - 20 - 3x - 3 = 24$$

$$x = 47$$

Q 2

Solve the equation $|2x+1| = 5x-3$.

ACE

$|2x+1| = 5x-3$

$$\begin{array}{ll} 2x+1 = 5x-3 & \text{and} \quad -(2x+1) = 5x-3 \\ 3x = 4 & -2x-1 = 5x-3 \\ x = \frac{4}{3} & 7x = 2 \\ & x = \frac{2}{7} \end{array}$$

Q 3

What is the solution to the equation $4^x - 9(2^x) + 8 = 0$?

ACE

- (A) $x=1$ and $x=3$
 (B) $x=1$ and $x=4$
 (C) $x=0$ and $x=3$
 (D) $x=0$ and $x=4$

Let $m = 2^x$
 then $m^2 - 9m + 8 = 0$
 $(m-8)(m-1) = 0$
 $m = 8 \quad m = 1$
 $2^x = 8 \quad 2^x = 1$
 $x = 3 \quad x = 0$

Q 4

Solve the following pair of simultaneous equations.

$$\begin{array}{l} x + y = 1 \\ 5x - 4y = 14 \end{array}$$

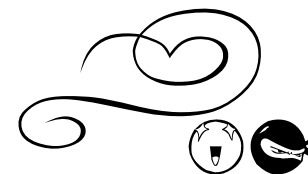
ACE

$$\begin{array}{ll} x + y = 1 & (1) \\ 5x - 4y = 14 & (2) \\ \text{Equation (1)} \times 4 & \\ 4x + 4y = 4 & (3) \\ \text{Equation (2)} + (3) & \\ 9x = 18 & \\ x = 2 & \\ \text{Substitute } x = 2 \text{ into equation (1)} & \\ 2 + y = 1 & \\ y = -1 & \\ \text{Solution is } x = 2 \text{ and } y = -1. & \end{array}$$



BASIC ARITHMETIC

PART D: EQUATIONS



Q 5

Solve $4^x - 5(2^x) + 4 = 0$

ACE

$$4^x - 5(2^x) + 4 = 0$$

Let $m = 2^x$

then $m^2 - 5m + 4 = 0$

$$(m-4)(m-1) = 0$$

$$m = 4 \qquad m = 1$$

$$2^x = 4 \qquad 2^x = 1$$

$$x = 2 \qquad x = 0$$

Q 6

Solve $x^4 - 5x^2 + 4 = 0$

BAULKHAM

$$\begin{aligned} x^4 - 5x^2 + 4 &= 0 \\ u^2 - 5u + 4 &= 0 \\ (u-4)(u-1) &= 0 \\ u=4 \text{ OR } u=1 \\ x^2=4 \text{ OR } x^2=1 \\ x=\pm 2 \text{ OR } x=\pm 1 \end{aligned}$$

Q 7

Solve $4x^2 + 4x - 3 = 0$ by the method of completing the square.

BAULKHAM

$$\begin{aligned} x^2 + x - \frac{3}{4} &= 0 \\ \left(x^2 + x - \frac{1}{4}\right) - \frac{3}{4} &= \frac{1}{4} \quad | \\ \left(x + \frac{1}{2}\right)^2 &= 1 \quad | \\ x &= -\frac{1}{2} \pm 1 \\ x &= -\frac{3}{2} \text{ OR } \frac{1}{2} \quad | \end{aligned}$$

Q 8

The solutions to the equation $2x^2 - 7x - 2 = 0$ are:

BAULKHAM

(A) $\frac{-7 \pm \sqrt{33}}{4}$

(C) $\frac{7 \pm \sqrt{33}}{4}$

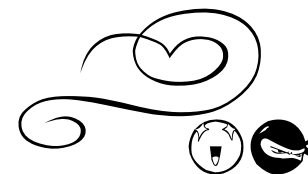
(B) $\frac{-7 \pm \sqrt{65}}{4}$

(D) $\frac{7 \pm \sqrt{65}}{4}$



BASIC ARITHMETIC

PART D: equations



Q 9

WESTERN

Solve the following equations simultaneously:

$$y = 2x + 4 \text{ and } y = x^2 + 1.$$

$$\begin{aligned} y &= 2x + 4 \quad \text{---(1)} \\ y &= x^2 + 1 \quad \text{---(2)} \end{aligned}$$

Sub (2) in (1)

$$\begin{aligned} x^2 + 1 &= 2x + 4 \\ x^2 - 2x - 3 &= 0 \\ (x - 3)(x + 1) &= 0 \\ x &= 3 \text{ or } x = -1 \end{aligned}$$

Therefore (3, 10) and (-1, 2)

Q10

BRIDIGINE

Solve the following simultaneous equations

$$x - y + 3 = 0 \quad \text{and} \quad xy = 10$$

$$\begin{aligned} x - y + 3 &= 0 \quad \left[x = \frac{10}{y} \right] \\ \frac{10}{y} - y + 3 &= 0 \quad \checkmark \\ 10 - y^2 + 3y &= 0 \\ 0 &= y^2 - 3y - 10 \\ 0 &= (y + 2)(y - 5) \\ y &= -2 \quad \checkmark \quad y = 5 \quad \checkmark \\ x &= -5 \quad \quad x = 2 \end{aligned}$$

Q11

$$\text{Solve } x^6 - 9x^3 + 8 = 0$$

DET

$$\begin{aligned} \text{Let } a &= x^3 \\ a^2 - 9a + 8 &= 0 \\ (a - 8)(a - 1) &= 0 \\ a &= 1 \text{ or } 8 \\ \therefore x^3 &= 1 \text{ or } x^3 = 8 \\ x &= 1, 2 \end{aligned}$$

Q12

FREEMAN

$$\text{Solve } x - \frac{2x - 4}{3} = 0$$

$$\begin{aligned} \frac{3x}{3} - \frac{2x - 4}{3} &= 0 \\ 3x - 2x + 4 &= 0 \\ x &= -4 \end{aligned}$$

Q13

FREEMAN

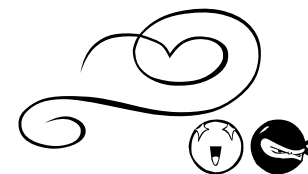
$$\text{Solve } 2x = 7 - \frac{5}{x}$$

$$\begin{aligned} 2x^2 &= 7x - 5 \\ 2x^2 - 7x + 5 &= 0 \\ (2x - 5)(x - 1) &= 0 \\ x &= \frac{5}{2}, 1 \end{aligned}$$



BASIC ARITHMETIC

PART D: equations



2unit

SELECTIVE

Q14

Solve simultaneously $x - y = 2$
 $3x + 2y = 1$

WESTERN

$$\begin{aligned} x - y &= 2 \quad \textcircled{1} \times 2 \\ 3x + 2y &= 1 \quad \textcircled{2} \\ 2x - 2y &= 4 \quad \textcircled{3} \\ 5x &= 5 \\ x &= 1 \\ 1 - y &= 2 \\ y &= -1 \quad (1, -1) \end{aligned}$$

Q15

Solve: $\frac{3}{x+2} = \frac{5}{2x-1}$

BRIDIGINE

$$\begin{aligned} \frac{3}{x+2} &= \frac{5}{2x-1} \\ 3(2x-1) &= 5(x+2) \\ 6x - 3 &= 5x + 10 \\ 6x - 5x &= 13 \quad \textcircled{3} \\ x &= 13 \end{aligned}$$

Q16

Solve using quadratic formula: $x^2 + 2 = -8x$

DET

$$\begin{aligned} x^2 + 2 &= -8x \\ x^2 + 8x + 2 &= 0 \\ x &= \frac{-8 \pm \sqrt{64 - 4 \times 1 \times 2}}{2} \\ &= \frac{-8 \pm \sqrt{64 - 8}}{2} = \frac{-8 \pm \sqrt{56}}{2} \\ &= \frac{-8 \pm \sqrt{14 \times 4}}{2} = \frac{-8 \pm 2\sqrt{14}}{2} \quad \textcircled{3} \\ &= \frac{2(-4 \pm \sqrt{14})}{2} = -4 \pm \sqrt{14} \end{aligned}$$

Q17

GIRAWEEEN

Solve the following pair of simultaneous equations.

$$\begin{aligned} 6x + 3y &= 0 \\ 2x - 5y &= 6 \end{aligned} \quad \begin{aligned} e) \quad 6x + 3y &= 0 \quad \textcircled{1} \\ 2x - 5y &= 6 \quad \textcircled{2} \times 3 \end{aligned}$$

$$\begin{aligned} 6x + 3y &= 0 \\ 6x - 15y &= 18 \\ -18y &= 18 \\ y &= -1 \end{aligned}$$

Sub $y = -1$ into (1)

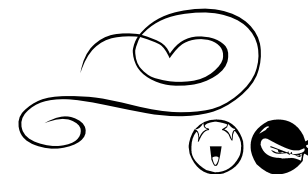
$$\begin{aligned} 6x + 3(-1) &= 0 \\ 6x - 3 &= 0 \\ 6x &= 3 \\ x &= \frac{1}{2} \end{aligned}$$

$\therefore x = \frac{1}{2}, y = -1$



BASIC ARITHMETIC

PART D: EQUATIONS



2unit

SELECTIVE

Q18

Solve the following pair of simultaneous equations:

$$7a + 3b = 36$$

$$5a + 2b = 25$$

GIRAWEEEN

$$\begin{array}{rcl} 7a + 3b = 36 & \text{--- (1)} & \times 2 \\ 5a + 2b = 25 & \text{--- (2)} & \times 3 \\ \hline 14a + 6b = 72 & \text{--- (3)} & \\ 15a + 6b = 75 & \text{--- (4)} & \\ \hline a = 3 & & \\ \text{Sub } a=3 \text{ into (1)} & & \\ 7(3) + 3b = 36 & & \\ b = 5 & & \\ \therefore a = 3, b = 5 & \text{(3)} & \end{array}$$

Q19

Solve: $|2x + 1| = 3x + 2$

BRIDIGINE

$$\begin{array}{l} \text{(11)} \quad |2x+1| = 3x+2 \\ \swarrow \quad \searrow \\ 2x+1 = 3x+2 \quad \text{OR} \quad 2x+1 = -(3x+2) \\ -1 = x \quad \quad \quad 2x+1 = -3x-2 \\ \quad \quad \quad 5x = -3 \\ \quad \quad \quad x = -\frac{3}{5} \end{array}$$

check solutions:

$x = -1$ $\times$ does not work

$x = -\frac{3}{5}$ $\checkmark$ works

$\therefore x = -\frac{3}{5}$ is the only solution

Q20

Solve the following equations simultaneously: $y = 2x + 4$ and $y = x^2 + 1$

DET

$$\begin{array}{l} c. \quad y = 2x + 4 \quad \text{--- (1)} \\ \quad y = x^2 + 1 \quad \text{--- (2)} \\ \text{Sub (1) into (2)} \\ 2x + 4 = x^2 + 1 \\ 0 = x^2 - 2x - 3 \\ 0 = (x-3)(x+1) \\ \therefore x = 3, -1 \\ \quad y = 10, 2 \end{array}$$

Q21

Solve $\frac{2w}{5} - \frac{w-1}{4} = w + 2$

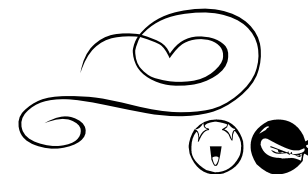
KINGS

$$\begin{array}{l} 8w - 5w + 5 = 20w + 40 \\ 3w + 5 = 20w + 40 \\ -17w = 35 \\ w = -\frac{35}{17} \end{array}$$



BASIC ARITHMETIC

PART D: EQUATIONS



Unit

SELECTIVE

Q22

Solve the following pair of simultaneous equations:

KINGS

$$\begin{aligned}x + 2y &= 5 \\ y &= x^2 - 5\end{aligned}$$

$$\begin{aligned}x + 2(x^2 - 5) - 5 &= 0 \\ 2x^2 + x - 15 &= 0 \\ (2x - 5)(x + 3) &= 0 \\ x &= -3, \frac{5}{2} \\ \therefore x &= -3 \text{ and } y = 4 \\ \text{or} \\ x &= \frac{5}{2} \text{ and } y = \frac{5}{4}\end{aligned}$$

Q23

Solve the following quadratic equation, leaving your answer in simplest surd form:

KINGS

$$4x^2 - 8x + 1 = 0$$

$$\begin{aligned}x &= \frac{8 \pm \sqrt{64 - 4(4)(1)}}{8} \\ &= \frac{8 \pm \sqrt{48}}{8} \\ &= \frac{8 \pm 4\sqrt{3}}{8} \\ &= \frac{2 \pm \sqrt{3}}{2}\end{aligned}$$

Q24

Expand and simplify: $(x - 3)(x + 3) - (x + 3)^2$

KINGS

$$\begin{aligned}(x - 3)(x + 3) - (x + 3)^2 &= x^2 - 9 - x^2 - 6x - 9 \\ &= -6x - 18\end{aligned}$$

Q25

Solve: $|2x - 4| - |x + 1| = 6$

KINGS

$$f) |2x - 4| - |x + 1| = 6$$

Case I

$$\begin{aligned}2x - 4 - (x + 1) &= 6 \\ 2x - 4 - x - 1 &= 6 \\ x &= 11\end{aligned}$$

① break down to 4 cases
② 4 answers.
③ check's correctly.

Case III

$$\begin{aligned}2x - 4 - (x + 1) &= 6 \\ 2x - 4 + x + 1 &= 6 \\ 3x - 3 &= 6 \\ 3x &= 9 \\ x &= 3\end{aligned}$$

Check

$$\begin{aligned}\text{Case I} \quad 18 - 12 &= 6 \checkmark \\ \text{Case III} \quad 2 - 4 &\neq 6 \times\end{aligned}$$

Case II

$$\begin{aligned}-(2x - 4) - (x + 1) &= 6 \\ -2x + 4 - x - 1 &= 6 \\ -3x + 3 &= 6 \\ -3x &= 3 \\ x &= -1\end{aligned}$$

Case IV

$$\begin{aligned}-(2x - 4) - (x + 1) &= 6 \\ -2x + 4 + x + 1 &= 6 \\ -x + 5 &= 6 \\ -x &= 1 \\ x &= -1\end{aligned}$$

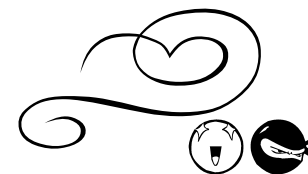
$$\begin{aligned}\text{Case II} \quad 6 - 0 &= 6 \checkmark \\ \text{Case IV} \quad 6 - 0 &\neq 6 \checkmark\end{aligned}$$

$$x = 11, \text{ or } -1$$



BASIC ARITHMETIC

PART D: EQUATIONS



Q26

Solve the equation $|2x - 3| = |5x - 2|$

ALL SAINTS

$$|2x - 3| = |5x - 2|$$

$$2x - 3 = 5x - 2$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

check :

$$LHS = \left| \frac{-2}{3} - 3 \right| = \frac{11}{3}$$

$$RHS = \left| \frac{-5}{3} - 2 \right| = \frac{11}{3}$$

$$\therefore x = -\frac{1}{3} \text{ valid}$$

$$(-1)(2x - 3) = 5x - 2$$

$$-2x + 3 = 5x - 2$$

$$5 = 7x$$

$$x = \frac{5}{7}$$

check :

$$LHS = \left| \frac{10}{7} - 3 \right| = \frac{11}{7}$$

$$RHS = \left| \frac{25}{7} - 2 \right| = \frac{11}{7}$$

$$\therefore x = \frac{5}{7} \text{ valid}$$

Q27

Solve for x: $\frac{1}{x-1} + \frac{x}{x+2} = 2$

ALL SAINTS

$$\frac{1}{x-1} + \frac{x}{x+2} = 2$$

$$\frac{x+2}{(x-1)(x+2)} + \frac{x(x-1)}{(x-1)(x+2)} = 2$$

$$\frac{x+2+x^2-x}{(x-1)(x+2)} = 2$$

$$x^2 + 2 = 2(x-1)(x+2)$$

$$x^2 + 2 = 2x^2 + 2x - 4$$

$$x^2 + 2x - 6 = 0$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-6)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{28}}{2}$$

$$x = -1 \pm \sqrt{7}$$

Q28

Show the equation $3x^2 = 9x - 5$ has solutions $\frac{9 \pm \sqrt{21}}{6}$

ALL SAINTS

$$3x^2 = 9x - 5$$

$$3x^2 - 9x + 5 = 0$$

$$x = \frac{9 \pm \sqrt{81 - 4 \times 3 \times 5}}{6}$$

$$= \frac{9 \pm \sqrt{21}}{6}$$

Q29

The weekly wage of 6 plumbers and 5 apprentices amount to \$6450 and the wages of 4 plumbers and 3 apprentices amount to \$4150. Find the weekly wage of a plumber and an apprentice.

ALL SAINTS

$$6P + 5A = 6450 \dots \dots \dots \text{eq \#1}$$

$$4P + 3A = 4150 \dots \dots \dots \text{eq \#2}$$

$$\text{eq \#1} \times 3$$

$$\text{eq \#2} \times 5$$

$$18P + 15A = 19350$$

$$- 20P + 15A = 20750$$

$$\text{-----}$$

$$-2P = -1400$$

$$P = 700$$

$$\text{-----}$$

$$6(700) + 5A = 6450$$

$$5A = 6450 - 4200$$

$$A = 450$$

The weekly wage of a Plumber is \$700 while his apprentice is \$450 .



BASIC ARITHMETIC

Part e: inequalities

2unit

SELECTIVE

Q 1

ACE

What values of x is $3x^2 \geq x$?

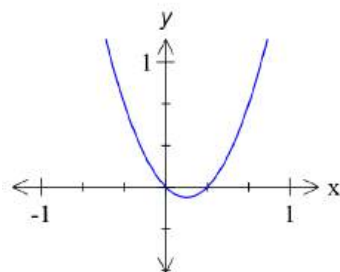
- (A) $x \leq 0$ or $x \geq \frac{1}{3}$
 (B) $x \geq 0$ and $x \leq \frac{1}{3}$
 (C) $x \leq 0$ or $x \geq -\frac{1}{3}$
 (D) $x \geq 0$ and $x \leq -\frac{1}{3}$

$$3x^2 \geq x$$

$$3x^2 - x \geq 0$$

$$x(3x-1) \geq 0$$

A



From the graph $x \leq 0$ or $x \geq \frac{1}{3}$

Q 2

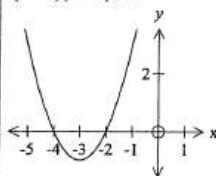
AMP

What is the solution to the inequation $x^2 + 6x + 8 \geq 0$?

- (A) $x \leq -2$ or $x \leq -4$
 (B) $x \leq -2$ or $x \geq -4$
 (C) $x \geq -2$ or $x \geq -4$
 (D) $x \geq -2$ or $x \leq -4$

$$x^2 + 6x + 8 \geq 0$$

$$(x+4)(x+2) \geq 0$$



D

Hence $x \geq -2$ or $x \leq -4$

Q 3

ASCHAM

What is the solution to the inequality $|2x-3| \geq 5$?

- (A) $x \geq 4$ or $x \leq -1$
 (B) $x \geq 4$ or $x \geq -1$ $|2x-3| \geq 5$
 (C) $x \leq 4$ or $x \leq -1$ $2x-3 \geq 5$ and $2x-3 \leq -5$
 (D) $x \leq 4$ or $x \geq -1$ $2x \geq 8$ $2x \leq -2$
 $x \geq 4$ $x \leq -1$

A

Q 4

ASCHAM

For what values of x is $x^2 \geq (x+1)(x+2)$

$$x^2 \geq (x+1)(x+2)$$

$$x^2 \geq x^2 + 3x + 2$$

$$0 \geq 3x + 2$$

$$3x \leq -2$$

$$x \leq -\frac{2}{3}$$



BASIC ARITHMETIC

PART E: INEQUALITIES

Unit

Selective

Q 5

Solve for x if

$$\left| \frac{4x+2}{5} \right| \leq 2$$

ASCHAM

$$-2 \leq \frac{4x+2}{5} \leq 2$$

$$-10 \leq 4x+2 \leq 10$$

$$-12 \leq 4x \leq 8$$

$$-3 \leq x \leq 2$$

Test $-2 \leq \frac{0+2}{5} \leq 2$

$$-2 \leq \frac{2}{5} \leq 2 \checkmark$$

Q 6

Solve and graph on a number line:

$$|2x-1| \geq 3$$

GIRAWEEEN

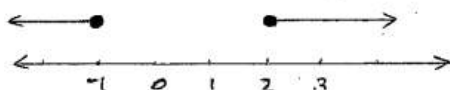
$$|2x-1| \geq 3$$

$$2x-1 \geq 3 \quad \text{or} \quad -(2x-1) \geq 3$$

$$2x-1 \geq 3 \quad \text{or} \quad 2x-1 \leq -3$$

$$2x \geq 4 \quad \text{or} \quad 2x \leq -2$$

$$\underline{x \geq 2} \quad \text{or} \quad \underline{x \leq -1}$$



Q 7

Solve the following quadratic inequation $x^2 + 3x \geq 10$

BAULKHAM

$$x \leq -5, x \geq 2$$

Q 8

Solve the following inequality.

$$-2 \leq 3x+7 < 10$$

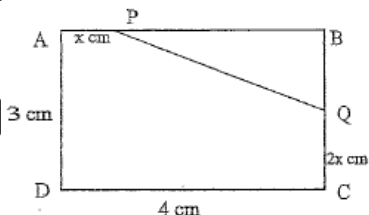
CRANBROOK

$$-9 \leq 3x < 3$$

$$-3 \leq x < 1 \checkmark$$

Q 1

ASCHAM



In the diagram above ABCD is a rectangle with $CD = 4$ cm and $AD = 3$ cm. P and Q are points on AB and BC respectively. If $AP = \frac{1}{2}QC = x$ cm.

- Show that the area of pentagon APQCD is given by the expression $6 + \frac{11x}{2} - x^2$
- Hence find the maximum area of the rectangle

$$\begin{aligned} \text{(i) Area of APQCD} &= \text{Area ABCD} - \text{Area } \triangle PBQ \\ &= (3 \times 4) - \frac{1}{2} \times (4-x)(3-2x) \\ &= 12 - \frac{1}{2}(12 - 8x - 3x + 2x^2) \\ &= 12 - \frac{1}{2}(12 - 11x + 2x^2) \\ &= 12 - 6 + \frac{11}{2}x - x^2 \\ &= 6 + \frac{11}{2}x - x^2 \end{aligned}$$

$$\begin{aligned} \text{(ii) } x &= \frac{-b}{2a} \\ &= \frac{-\frac{11}{2}}{-2} \\ &= \frac{11}{4} \end{aligned}$$

$\therefore$ Max when $x = \frac{11}{4}$ Since $a < 0$

Q 2

BAULKHAM

Make y the subject if $2\sqrt{x}\sqrt{y} - y + x^2 = x$

$$\begin{aligned} 2\sqrt{x}\sqrt{y} - y + x^2 &= x \\ y - 2\sqrt{x}\sqrt{y} + x &= x^2 \\ (\sqrt{y} - \sqrt{x})^2 &= x^2 \\ \sqrt{y} - \sqrt{x} &= \pm x \quad \text{OTHER METHODS} \\ \sqrt{y} &= \sqrt{x} \pm x \\ y &= (\sqrt{x} \pm x)^2 \quad \underline{2} \end{aligned}$$

Q 3

WESTERN

In a right angled triangle, one of the sides adjacent to the right angle is 6cm longer than the other. Find the length of all three sides of this triangle if its area is 216cm^2 .

Let x be one side, $\therefore$ other $= x + 6$

$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(x)(x + 6) = 216$$

$$\therefore x^2 + 6x = 432$$

$$x^2 + 6x - 432 = 0$$

$$(x - 18)(x + 24) = 0$$

$$x = 18 \text{ or } x = -24$$

Therefore $x = 18$

Three sides are 18, 24 and 30 by Pythagoras.

Q 4

The two shorter sides of a right-angled triangle have sides of lengths $2+\sqrt{3}$ and $2-\sqrt{3}$. Find the exact length of the hypotenuse.

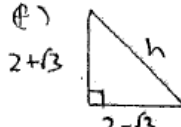
KAMBALA

$$h^2 = (2+\sqrt{3})^2 + (2-\sqrt{3})^2$$

$$= 4 + 4\sqrt{3} + 3 + 4 - 4\sqrt{3} + 3$$

$$h^2 = 14$$

$$h = \sqrt{14}$$



Q 5

Solve for real x . $x^2 - x + \frac{24}{x^2 - x} = 14$

MERIDAN

a) $x^2 - x + \frac{24}{x^2 - x} = 14$

Let $X = x^2 - x$

$X + \frac{24}{X} = 14$

$X^2 + 24 = 14X$

$X^2 - 14X + 24 = 0$

$(X-12)(X-2) = 0$

$X = 12, 2$

$\therefore x^2 - x = 12$ or $x^2 - x = 2$

$x^2 - x - 12 = 0$

$(x-4)(x+3) = 0$

$x = 4, -3$

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

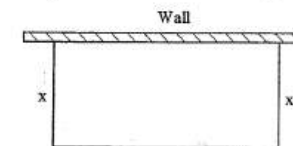
$x = 2, -1$

$x = -1, -3, 2, 4$

Q 6

A man wishes to make a rectangular garden using an existing wall as one side. He has 16 metres of wire netting.

- If the width of the garden is x metres, find the length and show that the area is given by $A = 16x - 2x^2$.
- Sketch $A = 16x - 2x^2$
- What is the maximum area of the garden?
- What dimensions give the maximum area?



$$2x + l = 16$$

$$l = 16 - 2x$$

$$A = l \times b$$

$$= (16 - 2x)(x)$$

$$= 16x - 2x^2$$

(ii) $y = 0, 16x - 2x^2 = 0$

$2x(8 - x) = 0$

$x = 0, 8$

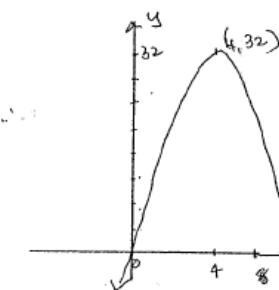
$$\text{Axis of symm. } x = \frac{-b}{2a}$$

$$= \frac{-16}{-4} = 4$$

$$\text{Vertex } x = 4$$

$$y = 16 \times 4 - 2 \times 4^2$$

$$= 32$$



(iii) Max = 32 m^2

$$(iv) x = 4$$

$$\text{width} = 4 \text{ m}$$

$$(\text{length}) l = 16 - 2 \times 4$$

$$= 8 \text{ m}$$

Q 7

The sides of a rectangle are given by $7-k$ and $7+k$. If the area of the rectangle is 36.75cm^2 , then its perimeter is equal to?

ALL SAINTS

(A) 14cm

(B) 28cm

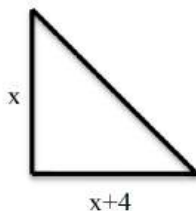
(C) 35cm

(D) 49cm

Q 8

In a right-angled triangle, one of the sides adjacent to the right angle is 4 cm longer than the other side. If the area of the triangle is 96 cm^2 , find the length of each of the three sides.

ALL SAINTS



$$\frac{1}{2}x(x+4) = 96$$

$$x^2 + 4x = 192$$

$$x^2 + 4x - 192 = 0$$

$$(x+16)(x-12) = 0$$

$$x = -16 \text{ or } 12$$

Since $x > 0$, $x = 12$

The sides are 12 cm, 16 cm and 20 cm.